Delayed Verification Destabilizes Multi-Agent LLM Belief: Instability Thresholds and Optimal Corrector Placement

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Abstract

Multi-agent large language model (LLM) systems often rely on verifier and critic agents to suppress hallucinations, but verification is delayed. During this delay, false claims can propagate through the agent network. We model this process as delayed consensus on a graph with grounded corrector nodes. Spectral decomposition by the grounded Laplacian yields a closed-form stability threshold for the verification dose: correction that is too strong or too delayed can turn consensus into oscillation. The most unstable regime occurs when the communication and verification delays coincide; for delay two, the threshold is the inverse golden ratio. The same framework gives a supermodular placement objective and a greedy $(1-1/e)$-approximation rule for assigning a limited corrector budget to influential nodes. Experiments across five open models confirm the predicted dose–delay oscillations. By contrast, grounded factual answering makes truth an *absorbing* boundary and eliminates the effect, suggesting that the instability is specific to signed-belief tasks while grounded verification remains stabilizing.

Keywords: multi-agent LLM systems; hallucination cascade; verification latency; delay stability; grounded Laplacian; leader selection; corrector placement.

1 Introduction

Large language models hallucinate, and in multi-agent systems hallucination stops being a static property of one output and becomes a dynamic process: claims are exchanged, revised, and reused as context, so an unsupported claim from one agent can be amplified by others [2, 4, 3]. The phenomenon is sharp even inside a single model: Zhang et al. show that once an LLM commits to a wrong answer it generates further false claims to justify it ("hallucination snowballing") even when it can separately recognize them as wrong [7]. A standard mitigation is to add verifier or critic agents that check claims against evidence and push the system back toward ground truth [8, 9, 10, 11].

Verification, however, has latency. A verifier reads a claim, retrieves evidence, and returns a correction only after several interaction steps; meanwhile the unverified claim has already propagated.

Delayed negative feedback is the classic ingredient of oscillation and instability in control systems, which raises a question the current LLM-agent literature does not ask: *can the very act of verification, if delayed, destabilize the factuality it is meant to protect, and how should correctors be dosed and placed to avoid this?*

The agent-safety literature is moving from post-hoc analysis toward online monitoring of cascades (causal cross-channel monitoring [20], online failure auditing [19], temporal-graph anomaly detection [21]), but these detect cascades; none models the closed-loop stability of the verification process itself, and none gives the detection-delay / dose trade-off that seven decades of quickest-change-detection theory provides [25, 26, 27, 30]. We supply the dynamics half of that missing piece (the closed-form dose and delay thresholds for the verification loop) with the complementary detection-delay versus false-alarm tradeoff treated for the single-agent signal in our companion work [36].

Our analysis instantiates, in the LLM verification loop, a delay-induced-instability framework recently developed for delayed institutional regulation of adaptive multi-agent systems [35], where a purely lagged alarm signal destabilizes an otherwise-stable equilibrium through a supercritical Hopf bifurcation. The closest dynamical accounts of LLM interaction are delay-free or single-agent: DeGroot-style consensus that converges monotonically [38], the average-consensus model of Chen et al. whose convergence is set by the graph spectrum [52], hidden-anchor deliberation [39], single-agent self-correction as feedback control with a static stability threshold [40], and the one empirical multi-agent factuality study [2]. None carries a verification *delay*; introducing it, and the oscillation it induces, is precisely our wedge.

Multi-agent debate is by now documented to degenerate, losing accuracy and failing to beat single-agent baselines [53]. What is missing is a *dynamical* account of that failure, which our threshold supplies.

This paper makes five contributions.

- **A model and a reduction.** We cast the corrected multi-agent loop as a delayed consensus with grounded correctors that the grounded Laplacian decouples into independent scalar delay recurrences (Lemma 1; Sections 3–4).
- **A closed-form verification-dose limit.** Above a critical correction strength the loop oscillates instead of converging, and this ceiling falls as the verification delay grows, reaching the inverse golden ratio at delay two (Theorem 1, Proposition 2, Lemma 2, Corollary 1; Section 4).
- **Optimal corrector placement.** Truth-tracking error is governed by a resolvent whose coherence is supermodular, so a greedy rule places a limited corrector budget within $1 - 1/e$ of optimal, concentrating on the network's amplifier and bridge nodes (Theorem 2; Section 5).
- **Two coupled delays.** The gossip and verification delays interact through a trinomial stability region whose worst case is the synchronized-delay corner, where the ceiling is again the inverse golden ratio (Theorem 3, Corollary 3; Section 6).
- **An empirical regime dichotomy.** The predicted signed dose–delay oscillation appears in real numeric-estimation debates across five open models, fixed *a priori* from $\beta_c(\delta)$ with no fitting. The same delay leaves grounded factual debate convergent, so the instability is native to signed belief, not to grounded factuality (Remark 1; Section 7).

Code and data reproducing every figure and result are available at `https://github.com/YehudaItkin/delayed-verification-llm`.

2 Related work

This paper sits between two literatures that rarely meet (the empirical study of reliability in multi-agent LLM systems and the control- and detection-theoretic study of delayed feedback) and the recurring gap is that the former offers no delay/stability guarantee while the latter has not been brought to bear on factuality verification.

On the LLM side, a fast-growing body of work characterizes how errors arise and spread between agents, but treats the process statically. Jamshidi et al. track claim-level inconsistency across agent chains and report a two-sided effect (deeper chains lower the explicit hallucination score yet erode factual accuracy [2]) with a companion model of collective hallucination mitigated by confidence-weighted aggregation and selective isolation [3]; relatedly, Xie et al. show that a single injected error can drive most frameworks to full "infection" unless provenance is tracked [4], and bias and unsafe content propagate the same way [5, 6]. Multi-agent debate can improve factuality [10, 11], but the same literature documents its failure modes (degeneration of thought and biased judges [12], gains attributable to majority voting rather than to debate itself [13], and accuracy that *decreases* over rounds through sycophantic consensus [14]) while the communication topology governs both accuracy and cost [15].

Closest in spirit is a nascent dynamical-systems line: DeGroot consensus whose disagreement decays at a rate set by the second graph eigenvalue [38], hidden-anchor deliberation that can escape the initial convex hull [39], and single-agent self-correction recast as feedback control with a static threshold [40]. All of these are delay-free or single-agent, so on the cooperative topologies they assume they converge monotonically or rest at a fixed threshold. Oscillation without delay is still possible by a separate route, signed or structurally unbalanced interaction, but that is a distinct mechanism from ours. Discrete delayed *averaging* is itself delay-robust: its convergence is set by the topology, not the delay magnitude [1]. The instability we study therefore originates in the delayed *corrector*, not the gossip; no prior model carries that verification *delay*, which is our wedge over the delay-free factuality recurrence of [2].

Finally, where we analyze the controller, a parallel effort builds *detectors*: post-hoc attribution of which step failed remains hard (11–41% step-localization for frontier models [16, 17, 18]), and the field is pivoting to online monitoring through trajectory-prefix auditing [19], cross-channel causal-influence monitoring of cascade onset [20], and temporal-graph anomaly detection [21].

On the control and detection side, the backbone we draw on is sequential change detection (the cumulative-sum statistic, CUSUM [25], the Shiryaev–Roberts and minimax theory [26, 27, 28, 29], and the standard surveys and monograph [30, 31, 32]) together with the discrete delay-stability result of Kuruklis [34] that our dose boundary specializes; the single-LLM signals that could feed such detectors, from semantic entropy [22] to SelfCheckGPT [23] and broader surveys [24], are by now mature. Only two LLM-side efforts touch this machinery directly, and neither targets multi-agent hallucination onset: [41] is the first to apply CUSUM to chain-of-thought, but to detect single-model confidence *convergence* for early exit, and [42] models interacting agents as Bayesian social learners with stochastic control to delay herding, without detection-delay bounds; the closest precursor is our own single-model formulation of hallucination *onset* as quickest change detection with Lorden delay bounds [36], which the present multi-agent, delayed-correction treatment lifts to a network.

The placement results we prove, in turn, are a port of classical network control: convergence-error coherence over the grounded Laplacian is supermodular, which yields greedy $(1-1/e)$ leader-selection guarantees [37, 44, 45, 47], whereas the only LLM-side analogue, Sherlock [43], places verifiers under a budget on a workflow graph through a counterfactual heuristic with no optimality guarantee and no stability analysis. The delay axis has a control precedent as well: Pirani et al. characterize the maximum-allowable-delay margin of leader–follower consensus through the grounded-Laplacian spectrum and turn it into a leader-placement centrality [46], but in continuous time and with a single delay; we add the discrete verification dose, the two coupled delays, and the LLM loop. The grounding signals themselves (claim verification and faithfulness [48, 49]) are standard.

3 Model

We abstract a round of multi-agent deliberation as follows. Each agent carries a scalar *belief* about a claim (its current confidence in the answer it would give), and the system advances in rounds: agents revise their belief toward those of their neighbours (debate), while designated *corrector* agents, which have access to verified evidence, push beliefs back toward the ground truth. Two features of deployed systems drive the dynamics and are the focus of this paper. First, correction is *delayed*: a verifier reads a claim, retrieves evidence, and replies only after a latency, so it acts on a stale belief. Second, some agents are *faulty*: they hold a fixed wrong belief that pulls their neighbours away from truth. The problem we study is then stated as: given the interaction graph, the verification delay, and the faulty forcing, (a) when does the loop drive every belief to the truth, and when does it instead oscillate, and (b) where should a limited budget of correctors be placed to keep the system on truth? The rest of this section makes the model precise; Sections 4–6 answer (a) and Section 5 answers (b).

Let $G = (V, E)$ be a graph on $V = \{1, \ldots, n\}$ with symmetric Laplacian $L = D - A \succeq 0$. Agent i holds a scalar belief $b_{i,t} \in \mathbb{R}$ about a claim with ground truth $b^\star$; the error is $e_{i,t} = b_{i,t} - b^\star$. A *corrector* set $R \subseteq V$ ($|R| = m$) consists of agents with access to verified evidence, held at truth ($e_{i,t} = 0$, $i \in R$), a discrete-time analogue of pinning control [51]. The remaining *free* agents $V_\mathrm{f} = V \setminus R$ ($n_\mathrm{f} = n - m$) carry error $e_t \in \mathbb{R}^{n_\mathrm{f}}$; a faulty subset injects a constant bias collected as a forcing $g \in \mathbb{R}^{n_\mathrm{f}}$. We take the Laplacian *symmetric*, meaning influence is reciprocal, as in peer debate or blackboard memory (the regimes of our experiments). This is the structural assumption behind the orthogonal decoupling of Lemma 1. Strictly directed pipelines (generator→verifier→rewriter) are non-normal and fall outside its scope. The belief is a single scalar and the corrector is *oracle-but-delayed*: content-level phenomena (truth dilution, claim provenance, epistemic laundering) are out of scope by construction, the destabilizing ingredient being the *delay* rather than verifier error.

Free agents run consensus (step $\eta > 0$) plus a uniform *delayed verification* of gain $\kappa > 0$ and integer latency $\delta \geq 1$:

$$e_{t+1} \;=\; (I - \eta L_\mathrm{g})\, e_t \;-\; \eta\kappa\, e_{t-\delta} \;+\; \eta g, \tag{1}$$

where $L_\mathrm{g} := L[V_\mathrm{f}, V_\mathrm{f}]$ is the principal submatrix of L on the free nodes (the *grounded Laplacian*). Reading (1) term by term: $(I - \eta L_\mathrm{g})\, e_t$ is the consensus (debate) step, in which each free agent moves toward its neighbours at rate η; $-\eta\kappa\, e_{t-\delta}$ is the correctors' restoring force toward truth, with *strength* κ and *delay* δ (it acts on the δ-step-old error); and ηg is the constant pull of the faulty agents. The verification strength κ and delay δ are the two control knobs, and everything that follows analyses (1) as a function of them.

For the loop to be well posed every free agent must be anchored, directly or through its neighbours, to some corrector.

Assumption 1 (grounding reachability)**.** Every connected component of the subgraph induced by V_f contains a node adjacent to R.

This is exactly the condition under which the grounded Laplacian is positive definite, $L_\mathrm{g} \succ 0$: the corrected loop then has a *unique* truth equilibrium and, in the absence of delay, contracts to it. A group of free agents cut off from every corrector would instead drift untethered, a zero mode of L_g that no verification dose can stabilize. We assume this throughout.

4 Stability and the verification dose

Lemma 1 (grounded-Laplacian decoupling)**.** *Let $L_\mathrm{g} = Q\,\mathrm{diag}(\mu_1, \dots, \mu_{n_\mathrm{f}})Q^\top$ $(0 < \mu_1 \le \cdots \le \mu_{n_\mathrm{f}})$. Put $x_t = Q^\top e_t$, $\hat{g} = Q^\top g$. Since κI is scalar (hence commutes with L_g),* (1) *decouples into independent scalar recurrences*

$$x_{i,t+1} = a_i\, x_{i,t} - \eta\kappa\, x_{i,t-\delta} + \eta\hat{g}_i, \qquad a_i \coloneqq 1 - \eta\mu_i. \tag{2}$$

The decoupling is the crux of the analysis. An n_f-dimensional *delayed* network, ordinarily an awkward object, collapses into independent scalar modes, so the entire stability question reduces to a single scalar delay recurrence $x_{t+1} - a_i x_t + \beta\, x_{t-\delta} = 0$ (with $\beta = \eta\kappa$), one copy per eigenvalue μ_i of L_g. We analyse this scalar recurrence once and then quantify the result over the spectrum.

Proposition 1 (explicit $\delta = 1$ region)**.** *For $\delta = 1$, mode* (2) *is asymptotically stable iff $\eta\kappa < 1$ and $\mu_i < 2/\eta + \kappa$.*

Proof. Characteristic $z^2 - a_i z + \eta\kappa$ with $a_i = 1 - \eta\mu_i$; Jury's conditions for $z^2 + c_1 z + c_0$ ($c_1 = -a_i$, $c_0 = \eta\kappa$) reduce to $\eta(\mu_i + \kappa) > 0$ (automatic), $\mu_i < 2/\eta + \kappa$, and $\eta\kappa < 1$. ☐

Theorem 1 (networked stability)**.** *Under Assumption 1, the homogeneous part of* (1) *is asymptotically stable iff $(a_i, \eta\kappa) \in \mathcal{D}_\delta$ for every $\mu_i \in \mathrm{spec}(L_\mathrm{g})$, where $\mathcal{D}_\delta$ is the unit-disk region of $z^{\delta+1} - az^\delta + \eta\kappa$. For $\delta = 1$ this is $\eta\kappa < 1$ and $\mu_{\max}(L_\mathrm{g}) < 2/\eta + \kappa$; the binding modes are the spectral extremes of L_g.*

Proof. By Lemma 1 the system is the direct sum of the scalar modes; it is stable iff each is. Apply Proposition 1. ☐

For $\delta = 1$ the loop is therefore benign: it is stable as soon as the verification gain stays below $1/\eta$ and the step size resolves the fastest grounded mode, so single-step verification cannot by itself destabilize the system. What follows shows that the *delay* breaks this guarantee. To see how, we track the *onset* of instability (the first parameter values at which a mode leaves the unit disk) for a general delay δ.

Proposition 2 (oscillatory boundary)**.** *Fix $\delta \ge 1$ and the mode $x_{t+1} - ax_t + \beta x_{t-\delta} = 0$ with $\beta = \eta\kappa > 0$, $|a| < 1$. As β grows from 0, the first root to reach the unit circle is a complex pair $e^{\pm i\theta}$, and the critical dose is*

$$a = \frac{\sin((\delta+1)\theta)}{\sin(\delta\theta)}, \qquad \beta_c = \frac{\sin\theta}{\sin(\delta\theta)}, \qquad \theta \in (0, \pi/\delta), \tag{3}$$

so $\kappa_{\max}(a,\delta) = \beta_c/\eta$; at the boundary the network oscillates at angular frequency $\omega = \theta_\star$ (period $2\pi/\theta_\star$).

Proof. See Appendix A. $\square$

The boundary (3) is opaque as written. Re-expressing it through Chebyshev polynomials makes both its closed form and its monotonicity transparent, and it is the monotonicity that turns the boundary into a usable design rule.

Lemma 2 (Chebyshev form; the dose is monotone)**.** *With $c = \cos\theta$ and $\sin(\delta\theta)/\sin\theta = U_{\delta-1}(c)$ (U = Chebyshev second kind), the boundary* (3) *is algebraic,*

$$a = \frac{U_\delta(c)}{U_{\delta-1}(c)}, \qquad \beta_c = \frac{1}{U_{\delta-1}(c)}, \qquad c \in \Big(\cos\tfrac{\pi}{\delta+1}, \cos\tfrac{\pi}{2\delta+1}\Big)\ (a \in (0,1)),$$

with $\beta_c(a,1) \equiv 1$, $\beta_c(0,\delta) \equiv 1$, and the exact value $\beta_c(a,2) = \frac{1}{2}(\sqrt{a^2+4} - a)$. The dose $\beta_c(a,\delta)$ is strictly decreasing in a on $(0,1)$, and strictly decreasing in δ (with $\beta_c < 1$ for all $\delta \geq 2$).

Proof. See Appendix B. $\square$

This monotonicity is what gives the dose limit its bite: because the ceiling falls as either the delay δ or the eigenvalue μ grows, the binding constraint is always a single, identifiable mode, which the corollary now pins down. Two remarks sharpen the claim. First, because (3) is a reparametrization of the exact, necessary-and-sufficient stability region of Kuruklis [34], $\kappa_{\max}(a,\delta)$ is not merely a sufficient bound but the *true* boundary; it is the discrete-time, delay-difference analogue of the continuous consensus delay margin $\tau^\star = \pi/(2\lambda_{\max})$ of Olfati-Saber and Murray [50]. Second, the integer δ is the operative case: the strength-direction monotonicity holds for all δ by induction (Appendix B), and the delay-direction is verified on the binding branch.

Corollary 1 (dose limit and binding mode)**.** *Assume $\eta\mu_{\max}(L_g) \leq 1$ (so each $a_i \in [0,1)$). Then the network dose limit is set by the* slowest *grounded mode, $\kappa < \kappa_{\max}\big(1 - \eta\mu_{\min}(L_g), \delta\big)$, and since grounding raises $\mu_{\min}(L_g)$ (Cauchy interlacing),* ***placement relaxes the delay-induced dose limit****. In particular: over-strong correction ($\kappa \geq \kappa_{\max}$) or over-delayed correction (larger δ, lowering $\kappa_{\max}$) destabilizes the loop into oscillation, a regime we call* verification-induced oscillation.

5 Optimal corrector placement

Given a budget of k correctors, *where* should they go? The average truth-tracking error is a supermodular function of the placement, so the one-pass greedy rule of Algorithm 1 is within a factor $1 - 1/e$ of optimal (Theorem 2) and concentrates correctors on the network's amplifier and bridge nodes. This is the actionable half of the title.

The delay drops out of the equilibrium of (1):

$$(L_g + \kappa I)\, e_\infty = g \quad \Longrightarrow \quad e_\infty(R) = (L_g + \kappa I)^{-1} g, \tag{4}$$

so *stability* is governed by $(\operatorname{spec} L_g, \kappa, \delta)$ while *truth-tracking error* is governed by the resolvent $(L_g + \kappa I)^{-1}$ acting on the fault forcing, the two decoupled levers of the same placement R.

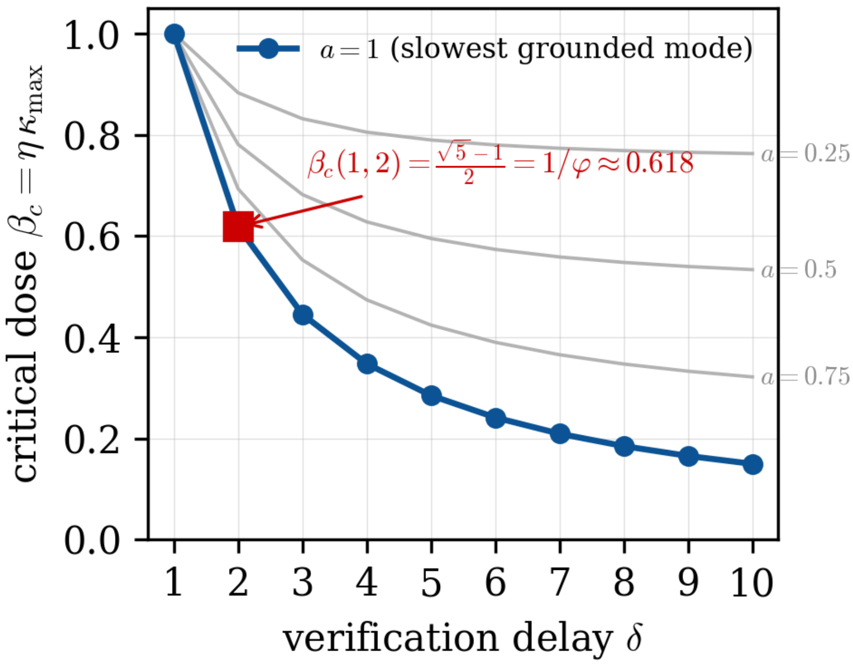

Figure 1: **The verification dose ceiling falls with delay.** Critical dose $\beta_c = \eta\kappa_{\max}$ versus verification delay δ for the binding mode $a = 1$ (blue) and three lighter modes (grey); the loop is stable below each curve. The ceiling decreases monotonically in δ and, at $\delta = 2$, equals the inverse golden ratio $(\sqrt{5}-1)/2 \approx 0.618$ (red). More verification *latency* therefore forces a strictly smaller safe verification *strength*: the dose–delay tradeoff that the rest of the paper exploits.

Corollary 2 (bounded steady-state error)**.** *Whenever the loop is stable, $e_t \to e_\infty$ and the truth-tracking error obeys the resolvent bound*

$$\|e_\infty\|_2 \;\le\; \frac{\|g\|_2}{\mu_{\min}(L_{\mathrm{g}}) + \kappa}, \tag{5}$$

independent of the step size η and the delay δ. Thus the same $\mu_{\min}(L_{\mathrm{g}})$ that the dose limit (Corollary 1) makes the binding stability constraint also controls the residual error, and, at fixed η, ***placing correctors never decreases $\mu_{\min}(L_{\mathrm{g}})$ (Cauchy interlacing), which both relaxes the stability limit and shrinks the error bound.***

Proof. At equilibrium (1) gives $(L_{\mathrm{g}} + \kappa I)e_\infty = g$; since $L_{\mathrm{g}} \succ 0$ is symmetric, $\|(L_{\mathrm{g}} + \kappa I)^{-1}\|_2 = 1/(\mu_{\min}(L_{\mathrm{g}}) + \kappa)$, and submultiplicativity gives (5). □

The bound (5) shows that pinning helps, but not *which* nodes to pin: it sees only $\mu_{\min}(L_{\mathrm{g}})$. Ranking nodes needs the full average-case error. Model a corrector at node i as a soft pin of weight $w > 0$ (hard grounding is $w \to \infty$), giving the steady-state operator

$$M(R) = L_{\mathrm{g}} + W_R + \kappa I, \qquad W_R = w \sum_{i \in R} e_i e_i^\top. \tag{6}$$

The fault location is unknown, so we average the forcing over it, $g \sim (0, \sigma^2 I)$; the expected truth-tracking energy $\sigma^2 \operatorname{tr} M(R)^{-2}$ then decreases monotonically with the *coherence* $H(R) = \operatorname{tr} M(R)^{-1}$, the standard submodular placement metric [44]. Choosing where to place the budget is therefore the cardinality-constrained problem

$$\min_{|R|=k} \; H(R) = \operatorname{tr} M(R)^{-1}, \tag{7}$$

the average truth-tracking error over where faults may strike.

The coherence is supermodular, so its reduction $\rho(R) = H(\varnothing) - H(R)$ is monotone and submodular, and a one-pass greedy rule solves (7) within a factor $1 - 1/e$ [47] (Theorem 2). The greedy step is cheap and interpretable, because its *marginal gain* has the Sherman–Morrison closed form

$$\Delta_i(R) := H(R) - H(R \cup \{i\}) = \frac{w \, \|M(R)^{-1} e_i\|^2}{1 + w \, e_i^\top M(R)^{-1} e_i} \;\geq\; 0, \tag{8}$$

a *resolvent centrality* that is largest at the *amplifier* and *bridge* nodes whose unverified error reaches the most of the network. Greedy pins the node of largest Δ_i, updates $M(R)^{-1}$ by one rank-one step, and repeats (Algorithm 1).

Algorithm 1: Greedy corrector placement for problem (7): spend the budget one node at a time, each step pinning the node of largest marginal gain.

Input: grounded Laplacian L_{g}, dose $\kappa > 0$, pin weight $w > 0$, budget k

Output: corrector set R with $|R| = k$

$R \leftarrow \varnothing$

for $j \leftarrow 1$ **to** k **do**

  foreach $i \notin R$ **do** $\Delta_i(R) \leftarrow w \, \|M(R)^{-1} e_i\|^2 \big/ \big(1 + w \, e_i^\top M(R)^{-1} e_i\big)$ // (8)

  $i^\star \leftarrow \arg\max_{i \notin R} \Delta_i(R)$ `// top amplifier / bridge node`

  $R \leftarrow R \cup \{i^\star\}$

return R

Theorem 2 (greedy placement is near-optimal)**.** *The reduction $\rho(R) = H(\varnothing) - H(R)$, equivalently the coherence $H(R) = \operatorname{tr} M(R)^{-1}$, is monotone non-decreasing and submodular with $\rho(\varnothing) = 0$. Hence the set R_{greedy} returned by the greedy rule satisfies, for every budget k,*

$$\rho(R_{\mathrm{greedy}}) \;\geq\; \Big(1 - \big(1 - \tfrac{1}{k}\big)^k\Big) \, \rho(R^\star) \;\geq\; \big(1 - \tfrac{1}{e}\big) \, \rho(R^\star), \tag{9}$$

where $R^\star \in \arg\max_{|R|=k} \rho(R)$ is an optimal size-k set. The bound is worst-case; in practice greedy is far closer to optimal (Fig. 2).

Proof. See Appendix D. □

On a graph of three 5-cliques chained by bridges (Fig. 2) the rule is well separated from naive placement: with $k = 8$ correctors greedy cuts the residual error $\operatorname{tr} M(R)^{-1}$ from 9.1 to 2.3, against 2.9 for degree-based and 2.5 for random placement, and its first picks are exactly the bridge and cluster-hub nodes. The supermodular structure and the $(1 - 1/e)$ guarantee are *inherited* from leader selection in linear multi-agent systems [37, 44] and hold for the diffusion model (4); carrying them to a fully nonlinear error-propagation map is an open assumption. Because grounding raises both $\mu_{\min}(L_{\mathrm{g}})$ (relaxing the dose) and $\mu_{\max}(L_{\mathrm{g}})$ (tightening the ceiling at fixed η), there is a finite optimal budget $k^\star$.

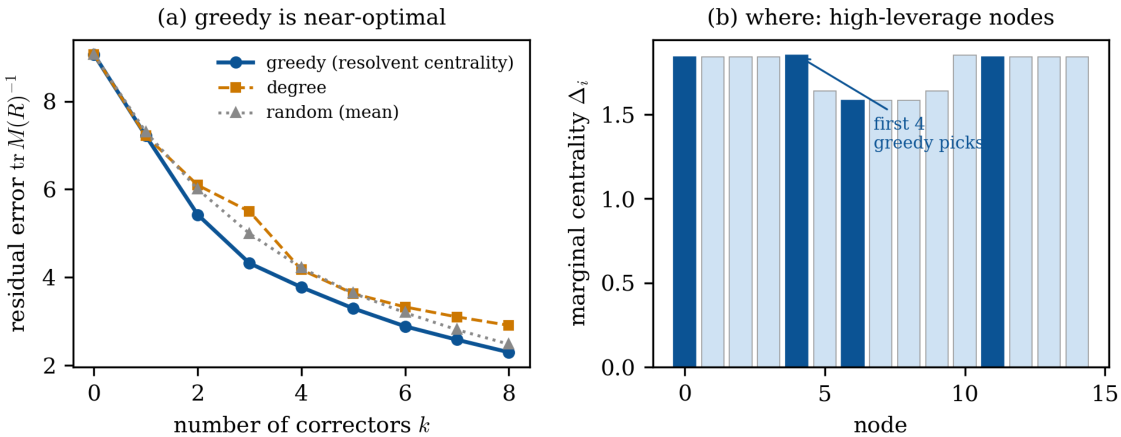

Figure 2: **Where to place correctors.** (a) On three 5-cliques chained by bridge edges, greedy selection by the resolvent centrality (8) lowers the residual error $\operatorname{tr} M(R)^{-1}$ faster than degree-based or random placement (mean over 300 orders), tracking the near-optimal frontier. (b) Marginal centrality Δ_i per node; the first greedy picks (dark) are the high-leverage bridge and hub nodes: the concrete answer to *where*.

6 Two coupled delays

Let gossip carry latency d and verification latency δ:

$$e_{t+1} = e_t - \eta L_{\mathrm{g}}\, e_{t-d} - \eta\kappa\, e_{t-\delta} + \eta g.$$

Lemma 1 still applies, giving per mode $x_{t+1} = x_t - \eta\mu\, x_{t-d} - \eta\kappa\, x_{t-\delta}$. A unit-circle root $\lambda = e^{i\theta}$ satisfies the pair

$$\cos\theta - 1 + \eta\mu\cos(d\theta) + \eta\kappa\cos(\delta\theta) = 0, \quad \sin\theta - \eta\mu\sin(d\theta) - \eta\kappa\sin(\delta\theta) = 0. \tag{10}$$

The stability of this two-delay trinomial is classical [33]; we recover its boundary by the D-decomposition method and read off the gossip-versus-verification specializations.

Theorem 3 (general two-delay boundary)**.** *For $d \neq \delta$, $p = \eta\mu$, $q = \eta\kappa$, the oscillatory boundary of* (10) *is*

$$p(\theta) = \frac{\sin(\delta\theta) - \sin((\delta+1)\theta)}{\sin((\delta-d)\theta)}, \qquad q(\theta) = \frac{\sin((d+1)\theta) - \sin(d\theta)}{\sin((\delta-d)\theta)}, \quad \theta \in (0,\pi),$$

and the stability region is the component of the origin bounded by this oscillatory curve together with the real-root line $p(-1)^d + q(-1)^\delta = 2$ (the $\lambda = -1$ crossing, which leaves the positive quadrant when d, δ are both odd). It degenerates correctly: at $d = 0$ to the single-delay boundary of Proposition 2, and as $d \to \delta$ (where $\sin((\delta - d)\theta) \to 0$) to the $a = 1$ corner of Corollary 3.

Proof. See Appendix C. □

Corollary 3 (synchronized delays are worst; a golden ratio)**.** *If $d = \delta$, the two terms merge into a single lag at $a = 1$, $x_{t+1} - x_t + \eta(\mu+\kappa)x_{t-\delta} = 0$, which is stable iff*

$$\eta(\mu+\kappa) < \beta_c(1,\delta) = \frac{\sin\frac{\pi}{2\delta+1}}{\sin\frac{\delta\pi}{2\delta+1}}, \qquad \textit{e.g.} \quad \beta_c(1,2) = \frac{\sqrt{5}-1}{2} = \frac{1}{\varphi}.$$

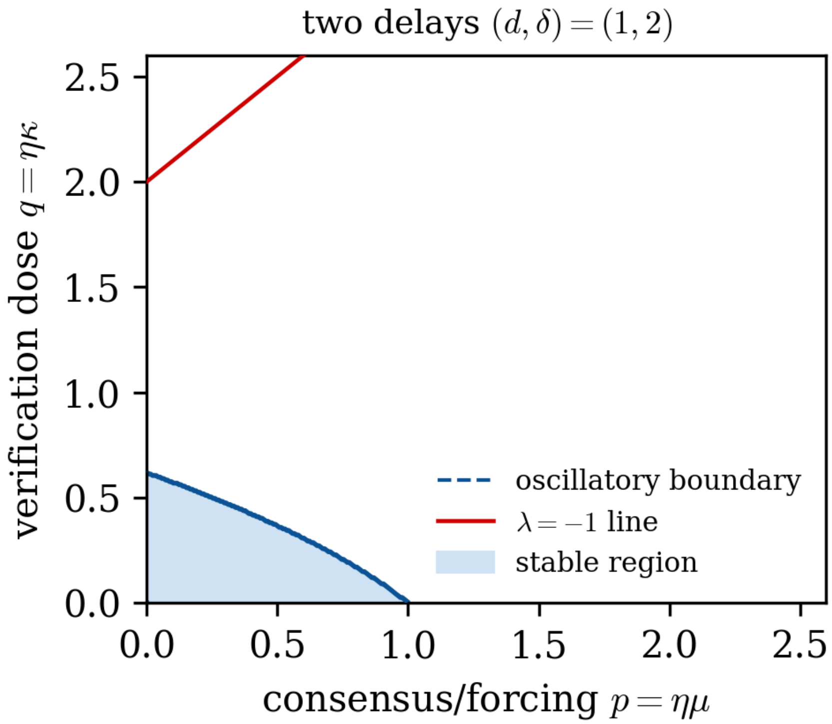

Figure 3: **A second delay shrinks the safe region.** Stability region (shaded) in the $(p, q) = (\eta\mu, \eta\kappa)$ plane for communication delay $d = 1$ and verification delay $\delta = 2$, bounded by the oscillatory boundary (Theorem 3, dashed) and the $\lambda = -1$ line (red, here non-binding). The dose ceiling on the q-axis is the same $1/\varphi \approx 0.618$ as in Fig. 1. Synchronizing the two delays ($d = \delta$) collapses the region the most (Corollary 3).

Since β_c decreases in a and $a = 1$ tops the branch, synchronizing communication and verification delays is the least stable configuration. Under Assumption 1 a single-delay mode has $a = 1 - \eta\mu_{\min} < 1$ strictly, so $a = 1$ is the limiting envelope (its $1/\varphi$ ceiling a supremum approached as $\mu_{\min} \to 0$); it is attained exactly *here, where both lags act at $a = 1$ by construction.*

7 Empirical validation

We ask three questions, each answered by one study below. RQ1: does the nonlinear loop lose stability exactly at the predicted dose limit $\kappa_{\max}(\delta)$? RQ2: in a real grounded factual debate, does verification *stabilize* the loop, and what happens when correction is too strong or ungrounded? RQ3: does the predicted signed dose–delay *oscillation* appear in real agents once the belief is a signed continuous quantity?

7.1 Onset at the predicted dose limit (RQ1)

We test whether the *linear* dose limit predicts onset in the *nonlinear* system with a saturating verifier,

$$e_{t+1} = (I - \eta L_{\mathrm{g}})e_t - \eta\kappa \tanh(e_{t-\delta}) + \eta g$$

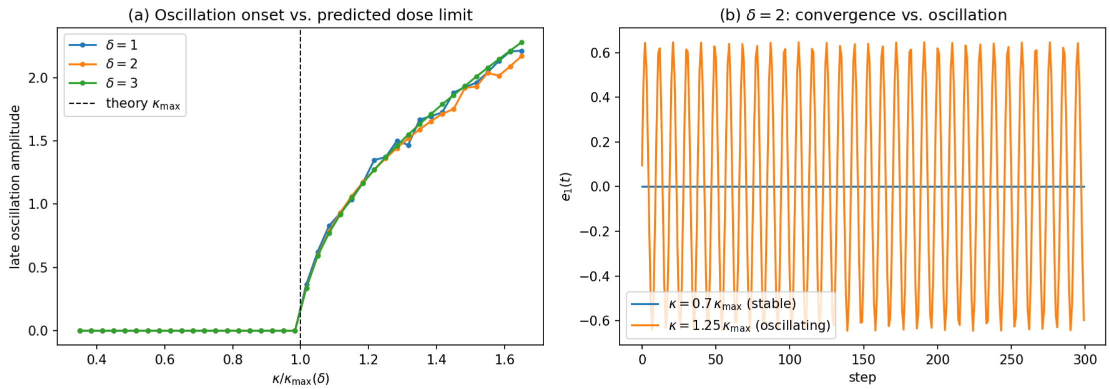

Figure 4: **Synthetic onset matches the predicted ceiling.** (a) oscillation amplitude collapses onto the predicted threshold $\kappa/\kappa_{\max} = 1$ for $\delta = 1, 2, 3$; (b) the δ=2 trajectory converges below the ceiling and oscillates above it.

($\tanh'(0) = 1$ matches the linearization), on a random grounded graph ($n_{\rm f} = 8$, η such that $\eta\mu \in (0, 1)$, a faulty node injecting a small bias). The onset of sustained oscillation $\kappa_{\rm crit}$ tracks the predicted ceiling $\kappa_{\max} = \beta_c(1 - \eta\mu_{\min}, \delta)/\eta$:

δ	$\kappa_{\max}$ (theory)	$\kappa_{\rm crit}$ (nonlinear)	ratio
1	16.30	16.57	1.017
2	10.21	10.38	1.017
3	7.45	7.57	1.017

The $\sim 2\%$ overshoot is the expected stabilizing effect of saturation; the δ=2 period (measured 9.38) matches $2\pi/\theta_\star = 9.72$. A real LLM-debate front-end substitutes for $\tanh(\cdot)$ without changing the analysis.

7.2 Grounded verification stabilizes the loop; delay alone does not (RQ2)

We now replace the synthetic surrogate by real LLM agents. As a reality check we instantiate the loop with a real debate among instances of a 35B reasoning model (Qwen3.6-35B) on factual questions it answers *incorrectly* when asked cold (the 30 PsiloQA questions on which it errs most when asked cold, with the Wikipedia passage as evidence): three free agents debate against a wrong majority (the forcing F) while a verifier (R) returns, with delay δ, a correction computed on the round-($t-\delta$) answer; per round we score each answer by its natural-language-inference (NLI) distance to gold. All statistics are paired at the question level (no pseudoreplication).

The robust, interpretable outcome is *convergence*, and on the theory's stable side it behaves as predicted. A grounded verifier of *moderate* strength drives the debate to truth (convergence ≈ 0.8); an *over-aggressive* verifier that forces agents to adopt its correction wholesale prevents convergence (≈ 0.4); and removing the grounding entirely, an ungrounded contrarian critic, destroys it (0 convergence, the majority answer flipping in 91% of rounds, a *debate collapse* of the kind documented for multi-agent debate [53]). Grounded correctors (Section 5) are what keep the loop convergent, the empirical counterpart of the stabilizing role they play in the theory. To test whether the *delay* itself destabilizes this grounded factual loop, we ran a pre-registered sweep over verification

Table 1: **The signed dose–delay oscillation replicates across five open models.** Fraction of debates whose signed error overshoots *through* zero (the Hopf signature) in the unstable cell (α=0.5, δ=6) against the stable cell (δ=1). All five reach 8/8 question-level unanimity (one-sided Wilcoxon floor p=0.004) and survive a Bonferroni correction.

Model	Developer	overshoot δ=6	overshoot δ=1
Qwen3.6-35B	Qwen	96%	0%
Qwen3-14B	Qwen	100%	0%
Mistral-7B	Mistral	100%	0%
Phi-4	Microsoft	96%	4%
Gemma-4-12B	Google	100%	0%

delay $\delta \in \{0, 1, 6\}$, fault forcing, and decoding temperature (2×180 debates over TriviaQA and PsiloQA). The consensus-error amplitude does *not* grow with δ: under a strong wrong majority it is already present at δ=0 (an instantaneous verifier), identifying it as forcing-driven churn rather than a delay cycle, and it vanishes once the majority is removed. The delay-induced oscillation is therefore confined to the signed-belief regime; grounded factual QA is stable in δ (Remark 1).

7.3 Signed dose–delay oscillation across models (RQ3)

To expose the signed oscillation directly we replace the factual answer by a *numeric estimate*: agents debate one of eight author-curated quantity questions with a known true value, so the signed error $e_t = (\bar{b}_t - b^\star)/\text{scale}$ can overshoot through zero, and the delayed corrector applies a *graded* relative correction of gain α on the round-($t{-}\delta$) estimate.[1] An agent following the corrector realizes the scalar delayed recurrence

$$e_{t+1} = e_t - \alpha\, e_{t-\delta}, \qquad \text{stable iff } \alpha < \beta_c(\delta) \quad (\beta_c(1){=}1, \quad \beta_c(6){\approx}0.24).$$

The prediction is fixed *a priori* from $\beta_c(\delta)$ with no fitting: α=0.5 should be stable at δ=1 but unstable at δ=6. It holds (Fig. 5), and the evidence is the *separation*, not a p-value. With the question as the unit (n=8, seeds averaged) the signed amplitude is $\sim 4.5\times$ larger at δ=6 than at δ=1 (0.27 vs 0.06). The signed overshoot through zero, the actual Hopf signature, occurs in 96% of (α=0.5, δ=6) runs against 0–4% at δ=1 (a single rephrasing-jitter crossing in Phi-4, none elsewhere).[2] Amplitude grows with both α and δ, and the only stable cell is small-α/small-δ, exactly the dose–delay region.

The result is not model-specific (Table 1): the same 8/8 unanimity and overshoot separation hold across five open models spanning four developers, survive a Bonferroni correction, and are directionally present in a 12B Mistral-Nemo (58% overshoot, p=0.07).

The instability does require agents that *follow* the correction — when they instead exert strong independent judgment they damp it, so deployed agents are if anything more stable than the linear worst case. The agents *follow* the recurrence without *copying* it: the pure linear map diverges (tail

[1] A signed coordinate is needed because the NLI distance used in RQ2 is non-negative, so it cannot exhibit the signed overshoot that *is* the oscillation; a sign-change count on such a magnitude trajectory is also confounded by answer-rephrasing noise and by a period-versus-window artifact, whereby a shorter delay mechanically produces more crossings.

[2] The one-sided Wilcoxon attains its n=8 floor $p = 1/2^8$=0.004, i.e. all 8/8 questions move as predicted, so the p certifies *unanimity*, not effect size; amplitude is the magnitude.

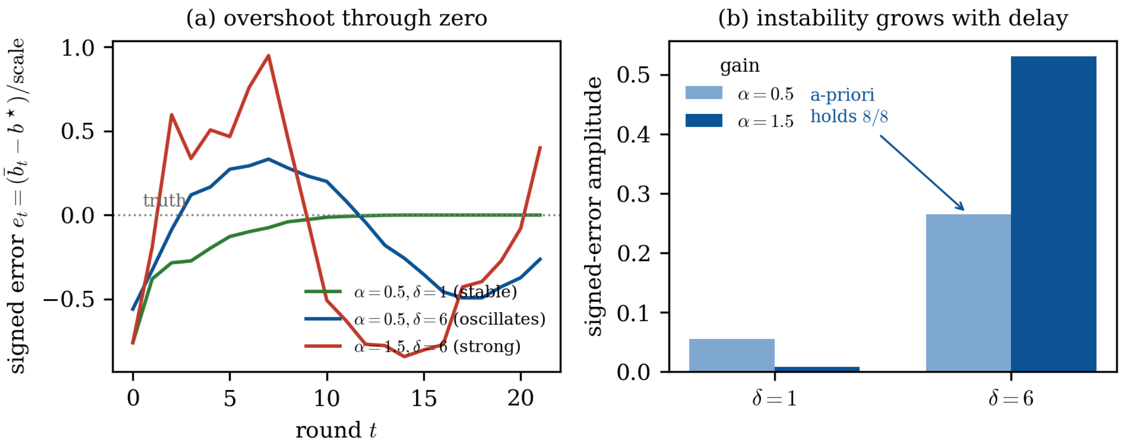

Figure 5: **Signed-error oscillation in a real Qwen3.6-35B numeric-estimation debate.** Agents debate a quantity with a known true value under a delayed relative correction of graded gain α; the signed error e_t can overshoot through zero. (a) Representative trajectories: the stable cell (α=0.5, δ=1) decays to truth without overshoot, while the delayed cells overshoot *through* zero and oscillate: the Hopf signature, present in 96% of (α=0.5, δ=6) runs versus $\leq 4\%$ at δ=1. (b) Signed-error amplitude grows with both delay and gain; the a-priori comparison amp(δ=6) > amp(δ=1) at α=0.5 holds (8/8 questions; the one-sided Wilcoxon hits its n=8 floor p=0.004). The only stable cell is small-gain/small-delay, exactly the dose–delay region of the theory.

amplitude $\sim$8, $|e|$ reaching 17) while the real debates saturate at bounded amplitude ($\lesssim$1), so the oscillation is an emergent response, not an arithmetic artifact of a hard-coded update.

The synthetic system, whose state genuinely *is* a signed error, remains the cleanest check of the *analytic threshold itself* (onset within 2%), a self-consistency test of the linearization, not an LLM validation.

8 Discussion

Verification is a control action, and like any delayed negative feedback it has a stability budget. Corollary 1 says a verifier can be *too* aggressive or *too* slow; Corollary 3 says synchronizing communication and verification latencies is the worst design. The practical reading is counterintuitive for the cascade literature, which treats more verification as strictly better: there is an optimal dose and an optimal placement, and the binding constraint is the slowest grounded mode, which placement can relax. This complements online cascade *detectors* [20, 19, 21] with a *controller*-side stability guarantee that detection alone does not provide.

Remark 1 (grounding removes the delay-induced oscillation)**.** The debates of Section 7 oscillate in the numeric-estimation regime yet not in grounded factual QA, and the reason is the state space. The instability is a property of the *signed* dynamics $x_{t+1} = x_t - \alpha\,\sigma(x_{t-\delta})$ (σ saturating): for $\alpha > \beta_c(\delta)$ the truth equilibrium is an unstable focus and the state runs a bounded limit cycle that crosses *through* zero. Grounded factual verification instead lives on a *non-negative* distance with truth as an *absorbing* boundary, $p_{t+1} = \max\big(0,\, p_t - \alpha\,\sigma(p_{t-\delta}) + g\big)$: a signed cycle that would overshoot truth is clipped at the boundary, so the delay-induced overshoot has nowhere to live.

The instability is therefore native to unconstrained signed belief and absent in grounded factual QA, where grounding both relaxes the dose limit, by raising $\mu_{\min}(L_{\mathrm{g}})$, and imposes the absorbing boundary that rules out the limit cycle.

9 Conclusion

We modeled the multi-agent LLM verifier loop as a delayed consensus over a graph with grounded corrector nodes, and derived closed-form thresholds showing that verification carries a stability budget. Correction that is too strong or too delayed destabilizes factual consensus into oscillation. The worst case is synchronizing the communication and verification delays, where the ceiling is the inverse golden ratio at delay two. A limited corrector budget is best placed greedily on high-influence nodes.

The empirics confirm the theory and bound its scope. A synthetic loop matches the predicted onset to within 2%. An a-priori, theory-derived experiment then reproduces the signed dose–delay oscillation in real LLM debates across five open models, with overshoot through truth in 96–100% of unstable runs versus 0–4% when stable. The same model explains why pure factual question answering does *not* oscillate: truth acts as an absorbing boundary, so the instability is native to signed-belief tasks while grounded factual verification is stabilizing.

The message reverses the field's default that more verification is always better: there is an optimal verification dose and an optimal place to apply it, and the binding constraint is the slowest grounded mode, which placement can relax.

10 Limitations

Several limitations bound the scope. The analysis is linear (local) around the truth-consensus; the nonlinear validation (Section 7) supports the local prediction but a global/contraction analysis is open. Unlike the dose limit, the placement rule is so far validated only on the linear surrogate, not with LLM agents. We model soft fixed-bias faults and node-level (state-overwriting) correctors; Byzantine faults and edge-level (message-filtering) correctors are future work, and the closed-form selection of which sub-arc of Theorem 3 bounds the origin component (the boundary curves themselves, the oscillatory curve and the $\lambda = -1$ line, are settled) is a minor remaining detail. The eigen-decoupling assumes a *symmetric* (reciprocal) interaction graph. Strictly directed topologies give a non-normal grounded Laplacian whose modes do not orthogonally separate, and transient growth can then precede the asymptotic threshold. The symmetric thresholds therefore need not stay conservative there, and a real-Schur/pseudospectral treatment is open.

The empirical support is of two kinds and should not be conflated. The signed numeric-estimation test (Section 7) is fixed *a priori* from theory and replicates across model families, but its unit is the question (n=8, seeds averaged), so its strength is the near-deterministic overshoot separation (96–100% versus 0–4%) rather than a large-sample effect size. The grounded factual-QA study is weaker as a *quantitative* link: the strong-κ delay effect is only marginal (Wilcoxon p=0.06) and the $\kappa\times\delta$ interaction is not significant (p=0.47), and the delayed autoregressive (AR) map identified from the logs recovers the linear coefficients only weakly (precision 1.0, recall 0.24), consistent with the absorbing boundary (Remark 1) damping the very oscillation that test looks for. A larger numeric study and a tighter identification of the LLM update map onto the linear coefficients are the key next steps.

Finally, the same grounded-Laplacian reduction should extend to heterogeneous correction gains, stochastic delays, and the coupling of this controller to an online change detector, closing the loop between *detecting* a hallucination cascade and *stably correcting* it.

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Appendix A Proof of Proposition 2 (oscillatory boundary)

Set $\lambda = e^{i\theta}$ in the characteristic equation,

$$e^{i(\delta+1)\theta} - a\, e^{i\delta\theta} + \beta = 0.$$

Separating imaginary and real parts gives the stability boundary in parametric form,

$$a(\theta) = \frac{\sin((\delta+1)\theta)}{\sin(\delta\theta)}, \qquad \beta(\theta) = a\cos(\delta\theta) - \cos((\delta+1)\theta) = \frac{\sin\theta}{\sin(\delta\theta)}.$$

A root can leave the unit disk in two ways. At $\lambda = +1$, $\beta = a - 1 \le 0$; at $\lambda = -1$, $\beta = (-1)^{\delta}(a+1)$, negative for odd δ but $a + 1 > 0$ for even δ. The complex-conjugate crossing instead carries

$$\beta_c = \frac{\sin\theta_\star}{\sin(\delta\theta_\star)},$$

and on the binding branch $\beta_c \le 1 < a + 1$ for $a \in (0, 1]$ (checked for $\delta \le 8$). Hence as β grows from 0 the complex pair reaches the unit circle first, including the even-δ case, where the real $\lambda = -1$ crossing at $\beta = a + 1$ comes strictly later.

Appendix B Proof of Lemma 2 (monotonicity of the dose)

Write $a_\delta(c) = U_\delta/U_{\delta-1}$. The Chebyshev recurrence $U_\delta = 2cU_{\delta-1} - U_{\delta-2}$ gives a continued fraction whose derivative telescopes,

$$a_\delta = 2c - \frac{1}{a_{\delta-1}}, \qquad a'_\delta = 2 + \frac{a'_{\delta-1}}{a^2_{\delta-1}}.$$

Since $a'_1 = 2$, induction gives $a'_\delta \ge 2$ wherever $U_{\delta-1} \ne 0$, so $\mathrm{d}a/\mathrm{d}c = a'_\delta > 0$. On the branch $c > \cos(\pi/\delta)$ (the largest zero of $U_{\delta-1}$), the $\delta - 2$ critical points of $U_{\delta-1}$ interlace its $\delta - 1$ zeros and so lie left of $\cos(\pi/\delta)$; thus $U_{\delta-1} > 0$ and $U'_{\delta-1} > 0$ there, and

$$\frac{\mathrm{d}\beta_c}{\mathrm{d}a} = -\frac{U'_{\delta-1}}{U^2_{\delta-1}\, a'_\delta} < 0,$$

the strict decrease in the correction strength a for *every* δ. For the integer delay, the trigonometric form on the binding branch is strictly decreasing in δ (verified for $\delta \le 8$):

$$\beta_c(1,\delta) = \frac{\sin\frac{\pi}{2\delta+1}}{\sin\frac{\delta\pi}{2\delta+1}} = 1,\ 0.618,\ 0.445,\ \ldots,\ 0.185 \qquad (\delta = 1,\ldots,8).$$

The boundary cases δ=1 and a=0 are exact, consistent with Kuruklis [34].

Appendix C Proof of Theorem 3 (general two-delay boundary)

Set $\lambda = e^{i\theta}$ in (10); it is linear in (p,q) with determinant $\sin((\delta - d)\theta)$, so Cramer's rule and the sum-to-product identities give the oscillatory boundary curve (the D-decomposition of the two-delay trinomial [33]). Evaluating the characteristic equation at $z = -1$ gives the real-root line

$$p\,(-1)^d + q\,(-1)^\delta = 2.$$

The two degenerations are the substitution $d = 0$ and the limit $d \to \delta$.

Appendix D Proof of Theorem 2 (greedy placement)

Throughout, write $M(R) = L_{\mathrm{g}} + \kappa I + w\sum_{i\in R} e_i e_i^\top \succ 0$ and $H(R) = \operatorname{tr} M(R)^{-1}$.

Monotonicity. For $i \notin R$, pinning is a positive-semidefinite update,

$$M(R\cup\{i\}) = M(R) + w\,e_i e_i^\top \succeq M(R) \implies M(R\cup\{i\})^{-1} \preceq M(R)^{-1},$$

so $H(R\cup\{i\}) \le H(R)$. Thus H is non-increasing and $\rho(R) = H(\varnothing) - H(R)$ is non-decreasing with $\rho(\varnothing) = 0$.

Marginal gain. By the Sherman–Morrison identity,

$$\left(M + w\,e_i e_i^\top\right)^{-1} = M^{-1} - \frac{w\,M^{-1} e_i e_i^\top M^{-1}}{1 + w\,e_i^\top M^{-1} e_i},$$

whose trace is $\operatorname{tr} M^{-1} - w\|M^{-1}e_i\|^2/(1 + w\,e_i^\top M^{-1} e_i)$. Subtracting recovers (8),

$$\Delta_i(R) = \frac{w\,\|M(R)^{-1} e_i\|^2}{1 + w\,e_i^\top M(R)^{-1} e_i} \ \ge\ 0.$$

Submodularity. We must show the marginal is non-increasing in the pinned set,

$$\Delta_i(R) \ge \Delta_i(S) \qquad \text{for } R \subseteq S,\ i \notin S.$$

Here the structure of L_{g} is essential. Because $M(R) = L_{\mathrm{g}} + \kappa I + W_R$ has nonpositive off-diagonal entries (from $-A$) and is positive definite, it is a symmetric nonsingular *M-matrix*, so

$$M(R)^{-1} \ge 0 \quad \text{entrywise},$$

and adding a nonnegative diagonal pin moves it entrywise toward $M(S)^{-1} \ge 0$. This entrywise monotonicity, not the Loewner ordering $M(S)^{-1} \preceq M(R)^{-1}$ alone (which does *not* imply the claim and indeed fails for general positive-definite M), forces Δ_i to be non-increasing in the set; equivalently, the coherence of a grounded Laplacian is supermodular under diagonal pinning [37, 44]. A numerical sweep confirms both halves (`placement_demo.py --sweep`): over 4000 random grounded Laplacians the marginal is non-increasing in 0/4000 trials, whereas dropping the M-matrix structure (arbitrary positive-definite M) violates it in 115/4000.

Greedy bound. For a non-negative monotone submodular ρ with $\rho(\varnothing) = 0$, the greedy rule of Algorithm 1, which adds the largest-Δ_i node at each step, attains the Nemhauser–Wolsey–Fisher guarantee [47],

$$\rho(R_{\text{greedy}}) \;\geq\; \left(1 - \tfrac{1}{e}\right) \rho(R^{\star}).$$

Appendix E Reproducibility

Every figure and reported number is reproducible from the accompanying code. Code and data: `https://github.com/YehudaItkin/delayed-verification-llm`.